

SAMINATHAN MS

Salem, India | +916385663777 | saminathanms2004@gmail.com | www.linkedin.com/in/SaminathanMS

Mechanical Engineering undergraduate (2025) with a focused interest in manufacturing systems, production processes, and mechatronic applications. Experienced in technical project execution, including the development of an **adaptive mechatronic lower limb exerciser for elderly wellness** and research on **methanol as a low reactivity fuel in RCCI combustion strategy**. Skilled in applying engineering principles, problem-solving, and teamwork to deliver efficient outcomes. Motivated to begin a career in process- and product-focused engineering roles within the manufacturing or production sector.

EDUCATION

College of Engineering Guindy, Anna University

B.E Mechanical Engineering – CGPA 7.69/10

Chennai, Tamil Nadu

10/2021 – 05/2025

Trinity Academy Matriculation Higher

Namakkal, Tamil Nadu

Secondary School

06/2020 – 05/2021

HSC – 92 %

Namakkal, Tamil Nadu

Trinity Academy Matriculation Higher

06/2018 – 03/2019

Secondary School

SSLC – 93 %

SKILLS

TECHNICAL SKILLS : Product Optimization ,Inventory management, Quality control and Experimental Analysis

SOFTWARES & TOOLS : MS – Excel, Canva , MS word, Canva, CAD (Intermediate), CATIA (Basics)

PERSONAL SKILLS : Leadership and team development, adaptability, analytical skills, Emotional Intelligence.

IMPLANT TRAINING DETAILS :

1. Steel Authority of India Limited, Salem – (07/2023).

Got practical Knowledge and witnessed how steels are made with various process that include Steel melting shop (SMS), Hot rolling mill, Cold rolling mill, and Blanking line.

PROJECTS DONE :

1. Investigating the effect of methanol as a low reactivity fuel in RCCI combustion strategy (12/2024)

This project explored the dual-fuel RCCI mode using methanol as a low reactivity fuel with diesel as the pilot fuel. Experimental results showed a 5–7% improvement in brake thermal efficiency, a 15–20% reduction in NOx emissions, and a significant decrease in soot/particulate matter compared to conventional diesel combustion. The study validated methanol's effectiveness in enhancing efficiency while reducing harmful emissions in IC engines

2. Adaptive Mechatronic Lower limb Excersicer (05/2025). (Team Lead)

This project focused on designing and fabricating a stainless-steel framed adaptive mechatronic exerciser for elderly rehabilitation. The prototype integrated **electrical muscle stimulation (EMS)** with an motorized mechanism to enable controlled exercise

WORKSHOPS :

1. IC – Engine workshop conducted at Ascension 21’ at Mechanical Engineering Department by Top Engineers. (02/2022).

EXPERIENCES :

1. CAREER GUIDANCE FORUM: PRESIDENT : (06/2024 –06/2025)

- Led an 50 plus member forum on career counselling ,alumni Engagement and Industry interaction.
- Organized an Mock Interview for 400 + Students with CEG alumni as Panelists
- Initiated student alumni Mentorship by connecting Student with alumni for Career Guidance
- Organized Workshop on Resume Building session enhancing students career readiness

2. CEG TECH FORUM

IR & MEDIA DOMAIN-Organizer (03/2023 –06/2024)

- Collected **CSR contributions and sponsorships** from companies to support flagship events
- Coordinated sponsorship negotiations and maintained relations with industry partners.
- Managed **media coverage and publicity** of events, ensuring broad student engagement.
- Collaborated with the media team on **content creation, promotion, and outreach.**

3. SOCIETY OF MECHANICAL ENGINEERS:

LOGISTICS DOMAIN : Co-Ordinator (09/2022 – 03/2023).

- Managed logistics for events like Pinnacle, Ascension, and workshops organized by the Society of Mechanical Engineers.
- Coordinated with vendors and handled material transport to ensure timely event setup.
- Led a 7-member team and used checklists and keep everything organized.

4. NATIONAL SPORTS ORGANISATION: Active volunteer in NSO Camp(2021-2023).

Had done immense service at Special Camp for junior students

HOBBIES :

- Reading books
- Playing Basketball
- Cycling.

LANGUAGES KNOWN :

- ENGLISH – Proficient (LRWS)
- TAMIL – Native (LRWS)